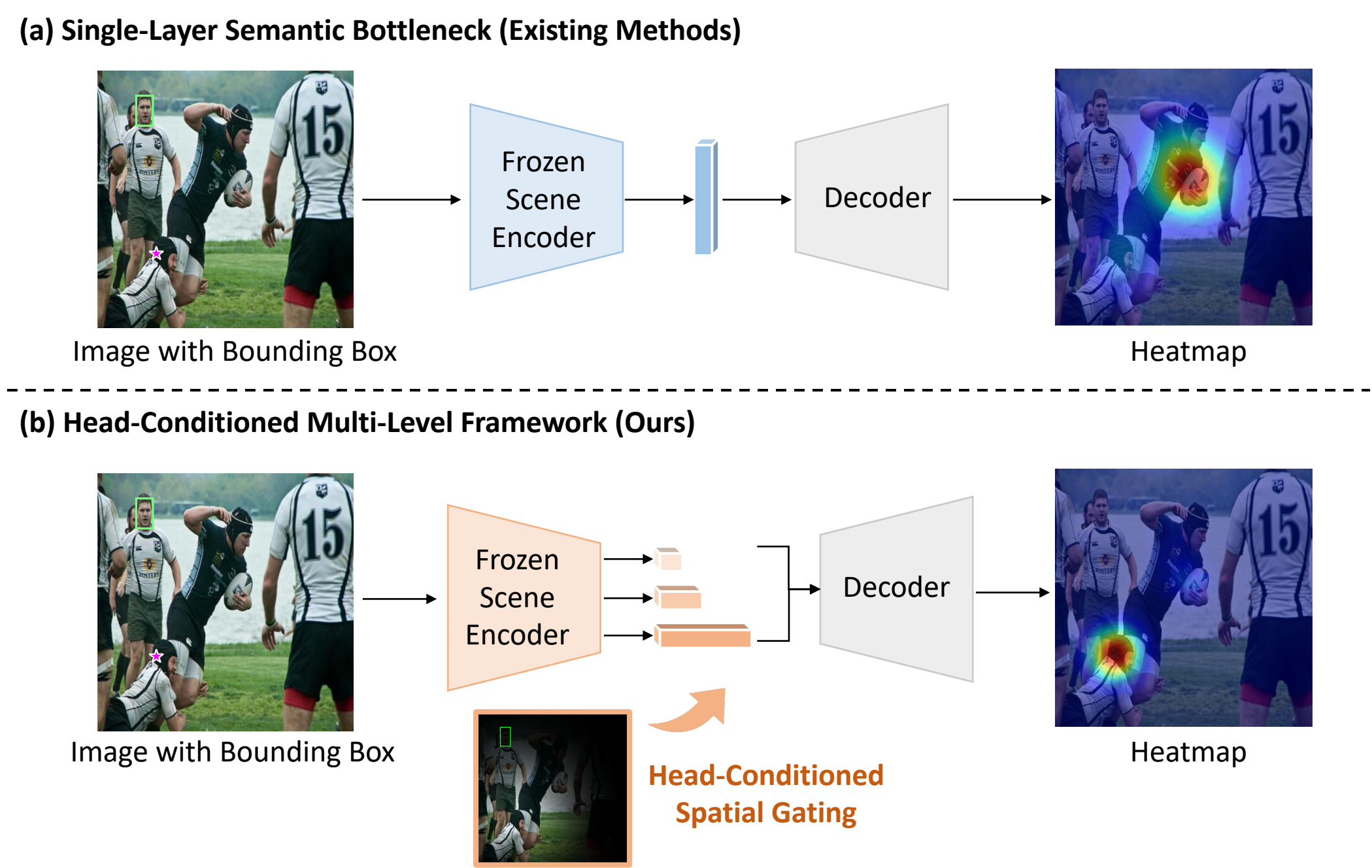

(a) Single-Layer Semantic Bottleneck (Existing Methods)
Frozen Scene Encoder
Decoder
Image with Bounding Box
Heatmap
(b) Head-Conditioned Multi-Level Framework (Ours)
Frozen Scene Encoder
Decoder
Image with Bounding Box
Head-Conditioned Spatial Gating
Heatmap